MATH0101 Linear Algebra for Data Science

Year:	2026–2027
Code:	MATH0101
Level:	5 (UG)
Normal student group(s):	UG: Y2 Mathematics programmes
Value:	15 credits (7.5 ECTS credits)
Term:	2
Assessment:	85% examination, 15% coursework
Normal Pre-requisites:	Grade A in A Level Mathematics
Lecturer:	Dr L Rila

Course Description and Objectives

The aim of this course is to provide an introduction to vectors, matrices, and least square methods, all basic topics in linear algebra, in the context of data science.

Recommended Texts

- S Boyd and L Vandenberghe, *Introduction to Applied Linear Algebra: Vectors, Matrices, and Least Squares* 3rd ed (Cambridge)
- G Strang, *Introduction to Linear Algebra* 5th ed (Wellesley–Cambridge)

Detailed Syllabus

Vectors: addition, scalar multiplication, inner product Clustering: norm, distances, clustering, the k-means algorithm. Linear independence: linear dependence, basis, orthonormal vectors. Matrix transformations: matrix operations, inverse matrices, matrix transformation in the plane, invariant points and invariant lines. Determinants: properties of determinants, co-factors and inverse matrices. Eigenvalues and eigenvectors: eigenvalues, eigenvectors, diagonalisation and symmetric matrices. Least squares: left and right inverses, pseudo-inverse, least square problem, least square data fitting. Complex numbers: operations, modules-argument form, exponential form, roots of complex numbers and geometric problems in the complex plane. Complex matrices: unitary and Hermitian matrices, eigenvalues, eigenvectors and diagonalisation

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